ADITYA BHOITE

bhoiteaditya20@gmail.com | +91-xxxxxxxxxx | Linkedin | GitHub

Education

Vidyalankar Institute of Technology, Mumbai 2022 – 2026
- Electronics & Computer Science | *CGPA*: **8.87**

PACE Junior Science College, Kalyan 2020 – 2022
- *HSC (Class XII)*

Skills

Languages: C++, Python, SQL, JavaScript
Libraries/Frameworks: NumPy, Pandas, TensorFlow, Scikit-learn, Matplotlib, Seaborn, Streamlit
Tools: Google Colab, Git/GitHub, Jupyter Notebook, Power BI, PostgreSQL, Docker, AWS

Certifications

- **Microwave Satellite Data Processing** – CSRE IITB

Work Experience

Machine Learning Intern – COE AI Lab, VJTI Mumbai June'25 – Jul'25
- Developed a **CNN**-based plant disease detection system using **TensorFlow/Keras**, achieving approximately **90% training accuracy and 88% validation accuracy** on the PlantVillage dataset.
- Evaluated model performance across training and validation data, analyzing classification behavior and generalization to improve the reliability of disease detection.
- Focused on building lightweight AI solutions suitable for use in resource-constrained agricultural environments.

Projects

US Labor Market Analytics Pipeline
End-to-end data pipeline analyzing US employment, wage, and AI-exposure trends across 830+ occupations spanning 11 years (2015–2025).
- Built an **ETL pipeline** integrating 4 data sources (BLS REST API, BLS bulk files, O*NET, LinkedIn postings), processing **8,900+ records** into a **PostgreSQL** database across 12 tables.
- Developed a **linear regression** employment forecasting model (2030–2035) and benchmarked it against official BLS projections across 755 occupations, quantifying a **34.3 percentage-point** average forecast deviation.
- Created a transparent **AI-exposure scoring model** using O*NET knowledge-area data and applied **K-Means clustering** to segment 755 occupations by wage, employment, and skill profile into 5 distinct labor-market tiers.
- Containerized the pipeline with **Docker** and deployed it on **AWS EC2** with **S3**-based data storage, and built an interactive dashboard surfacing wage, growth, and geographic insights.

Financial Resilience Intelligence
Analysis of financial resilience and vulnerability across 25,000+ US households using Federal Reserve survey data (2024–2025), from raw microdata to a deployed dashboard.
- Built a transparent financial resilience score from survey behavior (emergency fund, employment stability, debt stress, hardship exposure) and validated it against an independent Fed measure ($p < 0.001$, **Cohen's d = 1.48**).
- Derived financial-behavior features, improving **Logistic Regression ROC-AUC from 0.799 to 0.867**; benchmarked against **Random Forest** (0.891).
- Investigated a year-over-year resilience decline concentrated in low-income households (**31.5%** → **27.6%**),testing 4 explanations and identifying job instability and alternative-credit use as key drivers.
- Segmented households into **4 personas** using **K-Means**; deployed the analysis via **Streamlit**.

PlusOne – Sports Partner Matching Platform
A sports companion platform that helps users find players, join matches, and hire trainers, with an ML-based player recommendation system.
- Built a platform for browsing players, joining matches, and booking trainers based on rank, sport, and availability.
- Engineered player features (sport, rank, rating, win rate, availability) and used **cosine similarity** in **scikit-learn** to build a recommender that suggests compatible partners from player stats.